

Estimation by Confidence Interval

Interval Estimation

- An **interval estimate** provides a range likely to contain the parameter.
- The interval depends on the sample data and its sampling distribution.
- Increasing the sample size reduces variability, leading to a shorter, more precise interval.
- Interval estimates often provide more meaningful information than single-value estimates.

Estimation by Confidence Interval

Definition

A confidence interval (CI) estimate of the unknown population parameter θ is an interval, computed from sample data, that is likely to contain the true value of θ with a specified level of confidence.

For a 95% CI: "We are 95% confident that this interval contains the true population mean μ ."

- Does **not** mean there's a 95% probability μ is in the interval (since μ is fixed).
- Over many samples, 95% of computed intervals will contain μ .

What is a Confidence Interval?

- An interval estimate of a population parameter θ is an interval of the form:

$$\hat{\theta}_L < \theta < \hat{\theta}_U$$

where $\hat{\theta}_L$ and $\hat{\theta}_U$ depend on the value of the statistic $\hat{\theta}$ from a particular sample, and on the sampling distribution of $\hat{\theta}$.

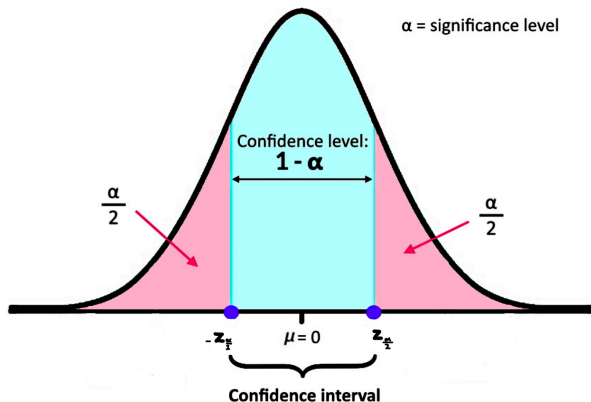
- Using the sampling distribution, we can choose $\hat{\theta}_L$ and $\hat{\theta}_U$ such that:

$$P(\hat{\theta}_L < \theta < \hat{\theta}_U) = 1 - \alpha$$

where α , known as **significance level**, represents the probability that the true parameter falls outside the confidence interval: the risk you're willing to accept for being wrong.

- The resulting interval is a $100(1 - \alpha)\%$ **confidence interval**.
- $1 - \alpha$ is the **confidence level**; the endpoints are the **confidence limits**.
- Common choices: $\alpha = 0.05$ (95% confidence), $\alpha = 0.01$ (99% confidence).
- Higher confidence (e.g., 99%) leads to wider intervals; lower confidence (e.g., 95%) yields narrower ones.
- Ideally, we want a short interval with high confidence.

What is a Confidence Interval?



Confidence Interval Estimate for Population Mean μ

Case (i): Normal Population with Known σ

Let a random sample $X_1, X_2, \dots, X_n$ of size n be drawn from a normal distribution with an unknown μ and known σ . Then the sampling distribution of the mean will be normal with a mean μ and a standard deviation $\sigma / \sqrt{n}$.

Standardized Z-Statistic

$$Z = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}} \sim N(0, 1)$$

Confidence Interval Inequality

$$P \left(-z_{\alpha/2} \leq \frac{\bar{X} - \mu}{\sigma / \sqrt{n}} \leq z_{\alpha/2} \right) = 1 - \alpha$$

Confidence Interval Estimate for Population Mean μ

Solve for μ

$$P\left(\bar{X} - z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X} + z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}\right) = 1 - \alpha$$

Hence, for a particular sample of size n , a $100(1 - \alpha)$ percent confidence interval for μ is given by:

Formula

$$\bar{x} \pm z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$$

- $\bar{x}$: sample mean
- $z_{\alpha/2}$: critical value from the standard normal distribution
- σ : population standard deviation
- n : sample size

Confidence Interval Estimate for Population Mean μ

Case (ii): Normal Population with unknown σ

Let a random sample $X_1, X_2, \dots, X_n$ of size n be drawn from a normal distribution with an unknown μ and unknown σ . We estimate population standard deviation σ by sample standard deviation S . For sufficiently large n , a $100(1 - \alpha)$ percent confidence interval for μ is given by:

Formula

$$\bar{x} \pm z_{\alpha/2} \cdot \frac{S}{\sqrt{n}}$$

- S : sample standard deviation

Confidence Interval Estimate for Population Mean μ

Case (iii): Non-normal Population with known and unknown σ (Large Samples)

Case (a): Known σ

Formula

$$\bar{x} \pm z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$$

Case (b): Unknown σ

Formula

$$\bar{x} \pm z_{\alpha/2} \cdot \frac{S}{\sqrt{n}}$$

Confidence Interval Estimate for Difference of Means

$$\mu_1 - \mu_2$$

Case (i): Non-normal Population with Known Standard Deviations

Let:

- Population 1 has mean μ_1 and standard deviation σ_1
- Population 2 has mean μ_2 and standard deviation σ_2
- Independent random samples of sizes n_1 and n_2 are taken:
 - Sample 1: $\bar{X}_1, S_1$
 - Sample 2: $\bar{X}_2, S_2$

Solve for $\mu_1 - \mu_2$

$$P\left((\bar{X}_1 - \bar{X}_2) - z_{\alpha/2} \cdot \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} \leq \mu_1 - \mu_2 \leq (\bar{X}_1 - \bar{X}_2) + z_{\alpha/2} \cdot \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}\right) = 1 - \alpha$$

Confidence Interval Estimate for Difference of Means

$$\mu_1 - \mu_2$$

Hence, for two independent samples, a $100(1 - \alpha)$ percent confidence interval for $\mu_1 - \mu_2$ is given by:

Formula

$$(\bar{x}_1 - \bar{x}_2) \pm z_{\alpha/2} \cdot \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

- $\bar{x}_1, \bar{x}_2$: sample means
- σ_1, σ_2 : population standard deviations
- n_1, n_2 : sample sizes

Confidence Interval Estimate for Difference of Means

$$\mu_1 - \mu_2$$

Case (ii): Normal Population with Unknown σ_1 and σ_2

Assume two independent random samples from normal distributions. For large sample sizes, we estimate σ_1 and σ_2 by S_1 and S_2 respectively. Then:

Formula

$$(\bar{x}_1 - \bar{x}_2) \pm z_{\alpha/2} \cdot \sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}$$

- S_1, S_2 : sample standard deviations

Confidence Interval Estimate for Difference of Means

$$\mu_1 - \mu_2$$

Case (iii): Non-normal Population with Known/Unknown σ (Large Samples)

Case (a): Known σ_1, σ_2

Formula

$$(\bar{x}_1 - \bar{x}_2) \pm z_{\alpha/2} \cdot \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

Case (b): Unknown σ_1, σ_2

Formula

$$(\bar{x}_1 - \bar{x}_2) \pm z_{\alpha/2} \cdot \sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}$$

Example: Confidence Interval for Population Mean

Problem:

A factory produces ball bearings whose lifespans are normally distributed. The population standard deviation is known to be 12 hours. A random sample of 25 ball bearings has an average lifespan of 200 hours. Construct a 95% confidence interval for the true mean lifespan of all ball bearings produced.

Solution

Given:

- Population standard deviation: $\sigma = 12$
- Sample size: $n = 25$
- Sample mean: $\bar{x} = 200$
- Confidence level: $95\% \Rightarrow \alpha = 0.05$
- $z_{\alpha/2} = z_{0.025} = 1.96$ (from standard normal table)

Confidence Interval Formula:

$$\bar{x} \pm z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$$

Substitute the values:

$$200 \pm 1.96 \cdot \frac{12}{\sqrt{25}} = 200 \pm 1.96 \cdot 2.4 = 200 \pm 4.704$$

$$\Rightarrow \text{Confidence Interval: } (195.296, 204.704)$$

We are 95% confident that the true mean lifespan of all ball bearings lies between **195.3 and 204.7 hours**.

Example: Confidence Interval for Difference in Means

Problem:

Two types of wire are being compared for durability. A random sample of 60 pieces of Type A wire shows an average breaking strength of 102.5 N with a standard deviation of 4.8 N. A separate random sample of 55 pieces of Type B wire has an average breaking strength of 98.1 N with a standard deviation of 5.2 N.

Construct a 99% confidence interval for the difference in population means $\mu_A - \mu_B$.

Solution

Given:

- Type A: $\bar{x}_1 = 102.5$, $S_1 = 4.8$, $n_1 = 60$
- Type B: $\bar{x}_2 = 98.1$, $S_2 = 5.2$, $n_2 = 55$
- Confidence level: 99% $\Rightarrow z_{\alpha/2} = 2.576$

CI formula:

$$(\bar{x}_1 - \bar{x}_2) \pm z_{\alpha/2} \cdot \sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}$$

Substitute values:

$$(102.5 - 98.1) \pm 2.576 \cdot \sqrt{\frac{4.8^2}{60} + \frac{5.2^2}{55}}$$

$$= 4.4 \pm 2.576 \cdot \sqrt{\frac{23.04}{60} + \frac{27.04}{55}} = 4.4 \pm 2.576 \cdot \sqrt{0.384 + 0.4925} = 4.4 \pm 2.576 \cdot \sqrt{0.8765}$$

$$= 4.4 \pm 2.576 \cdot 0.9362 \approx 4.4 \pm 2.412$$